

CBEH mock

Here is the fully updated 70-question mock exam, precisely calibrated to the **CHBE – Exam Template 2026** and the syllabus parameters. The interdisciplinary section has been updated to reflect the mixed-format style, and the specific sample questions provided in your template document (CHBE_template Exam 2026-3.pdf) have been seamlessly integrated into their respective modules.

As requested, the multiple-choice options have been strictly randomized to ensure absolutely no predictable patterns in the answer key.

CBEH Final Mock Examination 2026

Time Allowed: 90 Minutes

Total Questions: 70 (30 Cell Biology, 24 Histology, 12 Embryology, 4 Interdisciplinary)

Passing Requirement: Minimum 60% overall, with at least 50% in each module.

PART I: Cell Biology (30 Questions)

1. Multiple Choice: Which of the following transport mechanisms relies directly on the hydrolysis of ATP to move ions across the plasma membrane against their concentration gradient?

- A. Secondary active transport via symporters
- B. Facilitated diffusion via carrier proteins
- C. Primary active transport via the Na^+/K^+ pump
- D. Osmosis through aquaporins
- E. Ion channel-coupled receptor transport

2. True or False: Intermediate filaments exhibit rapid dynamic instability and treadmilling, making them the primary cytoskeletal element responsible for chromosome segregation during mitosis.

3. Multiple Choice: Which cellular compartment is primarily responsible for the sorting and modification of proteins destined for the lysosome, utilizing mannose-6-phosphate (M6P) tags?

- A. Early Endosome
- B. Smooth Endoplasmic Reticulum
- C. Peroxisome
- D. Trans-Golgi Network (TGN)
- E. Nucleolus

4. Open Question (Max 200 words): Briefly describe the structure of ribosomes, their function and localization in cells.

5. Multiple Choice: Which protein gives shape and stability to the nuclear envelope? A. Lamin B. Desmin C. Tubulin D. Actin E. Importin

6. Open Question (Max 200 words): Briefly describe the physico-chemical properties of the DNA double helix that allow it to be denatured (melted) and renatured (annealed). Mention the specific bonds involved.

7. Multiple Choice: The fundamental repeating unit of chromatin, consisting of approximately 147 base pairs of DNA wrapped around an octamer of core histones, is called a:

- A. Solenoid

- B. Chromatosome
- C. Telomere
- D. Centromere
- E. Nucleosome

8. True or False: Telomerase is a specialized reverse transcriptase that carries its own RNA template to solve the end-replication problem in eukaryotic linear chromosomes.

9. Multiple Choice: During eukaryotic transcription, which general transcription factor contains the TATA-binding protein (TBP) necessary for promoter recognition?

- A. TFIIB
- B. TFIID
- C. TFIIH
- D. TFIIIE
- E. TFIIA

10. Fill in the gap: The removal of introns and joining of exons to produce a mature mRNA transcript is catalyzed by a ribonucleoprotein complex known as the _____.

11. Multiple Choice: Which property of the genetic code explains why a single amino acid can be encoded by more than one synonymous codon?

- A. Universality
- B. Non-overlapping nature
- C. Redundancy (Degeneracy)
- D. Unambiguity
- E. Polarity

12. Matching: Match the chaperone/protein-folding concept with its correct description.

- 1. HSP70
- 2. HSP60 (Chaperonin)
- 3. Proteasome
- 4. Ubiquitin

- A. A small polypeptide tag that marks misfolded proteins for destruction.
- B. Forms a barrel-like isolation chamber for proteins to fold without aggregation.
- C. A multi-subunit protease complex that degrades tagged proteins.
- D. Binds to exposed hydrophobic regions of nascent polypeptides co-translationally.

13. Multiple Choice: Proteins destined for the nucleus are transported through nuclear pore complexes. What molecular signal is required for this specific targeting?

- A. Nuclear Localization Signal (NLS)
- B. KDEL sequence
- C. Mannose-6-phosphate
- D. N-terminal hydrophobic signal sequence
- E. Poly-A tail

14. True or False: In G-protein coupled receptor (GPCR) signaling, the activation of the receptor causes the alpha subunit of the G-protein to exchange GDP for GTP.

15. Multiple Choice: During which phase of mitosis do sister chromatids separate and begin to migrate toward opposite spindle poles?

- A. Prophase
- B. Metaphase
- C. Telophase
- D. Anaphase
- E. Prometaphase

16. Fill in the gap: The critical cell cycle checkpoint located at the G_1/S transition, which determines whether a cell commits to DNA replication in response to mitogens, is known as the _____ point.

17. Multiple Choice: Which of the following events is exclusive to Meiosis I and does NOT occur in Meiosis II or Mitosis?

- A. Condensation of chromatin into visible chromosomes
- B. Homologous recombination (crossing-over) at chiasmata
- C. Alignment of chromosomes at the metaphase plate
- D. Separation of sister chromatids
- E. Breakdown of the nuclear envelope

18. Open Question (Max 200 words): Differentiate between the intrinsic and extrinsic pathways of apoptosis, specifically naming the key initiating caspases for each.

19. Multiple Choice: Which of the following describes the structure of a retrovirus genome, such as HIV?

- A. Double-stranded DNA
- B. Single-stranded negative-sense RNA
- C. Double-stranded RNA
- D. Circular double-stranded DNA
- E. Single-stranded positive-sense RNA

20. True or False: Induced pluripotent stem cells (iPSCs) are derived by reprogramming adult somatic cells using a specific set of transcription factors (e.g., Yamanaka factors).

21. Multiple Choice: In the CRISPR-Cas9 gene editing system, what is the specific role of the Cas9 protein?

- A. It acts as an endonuclease that creates double-strand breaks at targeted loci.
- B. It acts as a reverse transcriptase to insert new DNA.
- C. It guides the RNA to the target promoter region.
- D. It ligates the broken DNA ends together via non-homologous end joining.
- E. It unwinds the DNA double helix like a helicase.

22. Fill in the gap: Unlike eukaryotic cells, prokaryotic cells lack a membrane-bound _____ to house their genetic material.

23. Multiple Choice: A frameshift mutation is most likely to result from which of the following alterations to the DNA sequence?

- A. A single base substitution from a purine to a pyrimidine
- B. A silent mutation in the third position of a codon
- C. An insertion or deletion of a number of nucleotides not divisible by three
- D. The precise deletion of exactly three consecutive nucleotides

E. A nonsense mutation

24. True or False: Prion diseases are caused by the transmission of misfolded proteins that induce normal versions of the same protein to adopt the pathological, aggregation-prone beta-sheet conformation.

25. Multiple Choice: Which second messenger is directly produced by the enzyme adenylyl cyclase upon GPCR activation?

- A. Inositol triphosphate (IP_3)
- B. Diacylglycerol (DAG)
- C. Calcium ions (Ca^{2+})
- D. Cyclic AMP (cAMP)
- E. Nitric oxide (NO)

26. Open Question (Max 200 words): Briefly explain the functional difference between a proto-oncogene and a tumor suppressor gene in the context of cancer progression.

27. Fill in the gap: In the intrinsic apoptotic pathway, the release of cytochrome c from mitochondria triggers the assembly of a multi-protein complex called the _____, which activates Caspase-9.

28. Multiple Choice: A point mutation that changes a codon specifying an amino acid into a premature STOP codon is classified as a:

- A. Missense mutation
- B. Silent mutation
- C. Splice-site mutation
- D. Frameshift mutation
- E. Nonsense mutation

29. True or False: Clathrin-mediated endocytosis is a receptor-independent, non-specific form of bulk transport (pinocytosis).

30. Multiple Choice: Which of the following best describes an "organoid"?

- A. A single layer of stem cells grown in a 2D petri dish.
- B. A 3D miniaturized and simplified in vitro version of an organ produced from stem cells.
- C. An isolated cellular organelle, such as a mitochondrion, maintained in vitro.
- D. A fully mature organ transplanted from an animal model into a human.
- E. A synthetic scaffold used for bone grafts.

PART II: Histology (24 Questions)

31. Multiple Choice: During standard histological tissue preparation, what is the primary purpose of the fixation step (e.g., using formalin)?

- A. To remove water from the tissue prior to embedding.
- B. To provide structural support for thin sectioning.
- C. To preserve tissue structure and prevent autolysis and putrefaction.
- D. To impart color contrast to the tissue structures for microscopy.
- E. To clear the lipids from adipose tissue.

32. Fill in the gap: The specific type of cell junction that forms an impermeable seal between adjacent epithelial cells, preventing paracellular diffusion, is called a _____ junction (zonula occludens).

33. Multiple Choice: A histological section reveals a single layer of cells that are taller than they are wide, with oval nuclei located near the basal domain. This epithelium is classified as:

- A. Simple squamous epithelium
- B. Stratified squamous epithelium
- C. Pseudostratified ciliated epithelium
- D. Transitional epithelium
- E. Simple columnar epithelium

34. True or False: Merocrine secretion involves the release of secretory products via exocytosis with no loss of cytoplasm or plasma membrane.

35. Multiple Choice: Which feature is uniquely characteristic of endocrine glands compared to exocrine glands?

- A. The presence of a highly branched duct system.
- B. The absence of ducts and the secretion of hormones directly into the bloodstream.
- C. An apical domain facing a free surface or lumen.
- D. The secretion of mucins and serous fluids.
- E. Their origin strictly from mesodermal tissue.

36. Open Question (Max 200 words): Describe the composition of the Extracellular Matrix (ECM) in connective tissue proper. Include both fibrous components and ground substance.

37. Multiple Choice: Which morphological feature strongly distinguishes brown adipose tissue from white adipose tissue?

- A. Multiple small lipid droplets and a high density of mitochondria.
- B. A single, large lipid droplet pushing the nucleus to the periphery.
- C. An extensive network of dense regular collagen fibers.
- D. The complete absence of vascularization.
- E. The presence of intercalated discs.

38. Fill in the gap: The specialized cells responsible for synthesizing the extracellular matrix of cartilage and residing within lacunae are called _____.

39. Multiple Choice: Identify the INCORRECT statement: A. During development, cartilage grows by interstitial and appositional mechanisms B. Volkman's canals connect osteocyte lacunae with haversian canals C. The osteoid is the unmineralized, organic portion of the bone matrix that forms prior the maturation of bone tissue D. The periosteum and endosteum contain osteoprogenitor cells E. The degradation activity of osteoclasts is stimulated by parathyroid hormone and inhibited by calcitonin

40. Multiple Choice: Which cell type is responsible for the production and secretion of insulin in the pancreas? A. Acinar cell B. Islet cell C. Alpha cell D. Beta cell E. Delta cell

41. Multiple Choice: In a lymph node, where are B-cell rich lymphoid follicles (nodules) primarily located?

- A. Medullary cords
- B. Paracortex
- C. Outer cortex
- D. Subcapsular sinus
- E. Trabeculae

42. Fill in the gap: The fundamental contractile unit of skeletal muscle, bounded by two Z-discs, is the _____.

43. Multiple Choice: Histologically, how is smooth muscle distinguishable from skeletal and cardiac muscle?

- A. It contains intercalated discs and branching fibers.
- B. It features peripheral, multiple nuclei within a single fiber.
- C. It is under strictly voluntary neural control.
- D. It lacks visible cross-striations and features central, spindle-shaped (cigar-shaped) nuclei.
- E. It lacks an extracellular matrix entirely.

44. True or False: In the peripheral nervous system (PNS), the myelin sheath is formed by oligodendrocytes.

45. Multiple Choice: Which glial cell type acts as the resident macrophage of the central nervous system (CNS), responsible for phagocytosis and immune defense?

- A. Microglia
- B. Astrocyte
- C. Ependymal cell
- D. Schwann cell
- E. Satellite cell

46. Matching: Match the specific lining epithelium with its typical location in the human body.

- 1. Simple squamous epithelium
- 2. Pseudostratified ciliated columnar epithelium
- 3. Transitional epithelium (Urothelium)
- 4. Stratified squamous non-keratinized epithelium

- A. *Urinary bladder*
- B. *Trachea and primary bronchi*
- C. *Esophagus and vagina*
- D. *Alveoli of the lungs and endothelium of blood vessels*

47. Multiple Choice: Which connective tissue type is characterized by densely packed, parallel bundles of type I collagen fibers and is typically found in tendons and ligaments?

- A. Dense irregular connective tissue
- B. Loose (areolar) connective tissue
- C. Reticular connective tissue
- D. Dense regular connective tissue
- E. Mucoid connective tissue

48. Fill in the gap: The process of bone formation where bone develops directly from mesenchymal tissue without a cartilage intermediate is called _____ ossification.

49. Multiple Choice: Which lymphoid organ is responsible for the maturation of T-lymphocytes and exhibits a distinct cortex and medulla, featuring Hassall's corpuscles in the medulla?

- A. Spleen
- B. Thymus

- C. Palatine tonsil
- D. Lymph node
- E. Appendix

50. True or False: Intercalated discs are unique structural features of skeletal muscle that facilitate synchronized, voluntary contractions.

51. Multiple Choice: The Blood-Brain Barrier (BBB) is primarily maintained by tight junctions between endothelial cells, supported by the end-feet of which glial cell?

- A. Oligodendrocytes
- B. Microglia
- C. Astrocytes
- D. Ependymal cells
- E. Schwann cells

52. Open Question (Max 200 words): Describe the histological structure and mechanism of the neuromuscular junction (motor endplate) during synaptic transmission.

53. Fill in the gap: The primary resident cell type of connective tissue proper, responsible for secreting collagen, elastin, and ground substance, is the _____.

54. Multiple Choice: Which microscopy technique utilizes a beam of electrons transmitted *through* an ultra-thin specimen to achieve extremely high-resolution images of internal cellular organelles?

- A. Scanning Electron Microscopy (SEM)
- B. Phase-contrast light microscopy
- C. Confocal fluorescence microscopy
- D. Dark-field microscopy
- E. Transmission Electron Microscopy (TEM)

PART III: Embryology (12 Questions)

55. Multiple Choice: The part of the sperm containing proteolytic enzymes to digest the zona pellucida is called: A. Capacitor B. Head C. Acrosome D. Cumulus E. Corona

56. Multiple Choice: A 55-year-old man has noted sharp pain in his right lower extremity for the past 2 months. MR imaging of his spine shows impingement on a spinal nerve root by a herniated structure located between L5 and S1. From which of the following embryonic derivatives does this herniated structure most likely arise? A. Notochord B. Amnion C. Cloaca D. Lateral Plate Mesoderm E. Neural crest

57. Fill in the gap: In the seminiferous tubules, the supporting cells that form the blood-testis barrier and nourish developing spermatozoa are called _____ cells.

58. True or False: Implantation of the human blastocyst typically occurs exactly 24 hours after fertilization.

59. Multiple Choice: During the third week of development, gastrulation results in the formation of:

- A. The bilaminar embryonic disc
- B. The neural tube
- C. The syncytiotrophoblast
- D. The three primary germ layers (ectoderm, mesoderm, endoderm)
- E. The extraembryonic mesoderm

60. Open Question (Max 200 words): Explain the structure and primary functions of the placental barrier (fetal-maternal interface). Mention the layers that separate fetal blood from maternal blood.

61. Multiple Choice: The process of embryonic folding in the fourth week converts the flat trilaminar disc into a cylinder. This folding directly leads to the incorporation of the yolk sac to form the:

- A. Neural tube
- B. Primitive gut tube
- C. Notochord
- D. Amniotic cavity
- E. Allantois

62. Fill in the gap: The morphogenetic process that leads to the formation of the neural tube from the neural plate is called _____.

63. Multiple Choice: Which of the following structures is derived from the embryonic *ectoderm*?

- A. The skeletal system
- B. The epithelial lining of the gastrointestinal tract
- C. The muscular system
- D. The dermis of the skin
- E. The central and peripheral nervous systems

64. True or False: The notochord provides the longitudinal axis of the embryo, induces the overlying ectoderm to form the neural plate, and eventually becomes the nucleus pulposus of the intervertebral discs.

65. Multiple Choice: The umbilical cord normally contains which of the following blood vessels?

- A. Two umbilical veins and one umbilical artery
- B. One umbilical vein and one umbilical artery
- C. Two umbilical arteries and one umbilical vein
- D. Two umbilical arteries and two umbilical veins
- E. Only lymphatic vessels

66. Matching: Match the embryonic structure with its ultimate developmental fate.

- 1. Inner Cell Mass (Embryoblast)
- 2. Trophoblast
- 3. Neural Crest Cells
- 4. Splanchnic (visceral) lateral plate mesoderm

- A. *Forms the wall of the primitive gut tube.*
- B. *Gives rise to the fetal portion of the placenta (chorion).*
- C. *Migrates to form peripheral ganglia, melanocytes, and craniofacial bones.*
- D. *Gives rise to the entire embryo proper.*

PART IV: Interdisciplinary (4 Questions)

67. Multiple Choice: Which of the following best explains why epithelial tissues can maintain rapid turnover and effective barrier function? A) Epithelial tissues are composed of multinucleated cells that divide asynchronously to maintain homeostasis. B) Epithelial cells exhibit high proliferative

capacity due to the presence of adult stem cells located in specific niches, and they are tightly joined by junctional complexes that maintain polarity and barrier integrity. C) Rapid turnover is sustained by fibroblasts that stimulate epithelial mitosis via collagen production. D) Epithelial regeneration relies mainly on apoptosis of surface cells, with minimal need for proliferation. E) Epithelial cells function independently of the extracellular matrix, allowing for higher renewal rates.

68. True or False Cluster: Evaluate the following statements regarding mitochondria across cell biology, histology, and embryology: A) Cardiac muscle cells have high mitochondrial density to sustain continuous ATP production. B) Transmission electron microscopy can be used to visualize mitochondrial cristae and matrix density in situ. C) Mitochondria in early embryos are maternally inherited and essential for energy supply during cleavage. D) Mitochondria cannot be detected by any histological staining techniques.

69. Open Question (Max 200 words): CRISPR-Cas9 genome editing has recently been applied to treat certain hematological diseases. Choose one such condition and briefly explain how CRISPR-Cas9 is used to address it. In your short answer refer to:

- Cell biology concepts such as gene expression, stem cell manipulation, or DNA repair mechanisms;
- Histological aspects including the structure and function of the affected blood cells or tissues.

70. Matching: Match the cell cycle or embryological stage with its defining cellular/histological outcome.

1. G1/S Transition (Restriction Point)
2. S Phase
3. Metaphase-to-Anaphase Transition
4. Cleavage

A. Rapid mitotic divisions of the zygote without intervening growth phases, producing a multicellular morula.

B. Activation of APC/C leads to securin and cyclin B degradation, separating sister chromatids.

C. Retinoblastoma (Rb) protein is phosphorylated by Cyclin-CDK complexes, committing the cell to division.

D. Centrosomes duplicate and the genome is replicated, yielding cells with a 4c DNA content.

ANSWER KEY

Part I: Cell Biology

1. **C**
2. **False** (Microtubules are responsible for chromosome segregation; IFs provide tensile structural support).
3. **D**
4. **Open Concept:** Ribosomes are large ribonucleoprotein complexes made of a large and small subunit (e.g., 60S/40S in eukaryotes), comprising rRNAs and proteins. They catalyze peptide

bond formation (translation) and are located free in the cytosol (for intracellular proteins) or bound to the rough ER (for secreted/membrane proteins).

5. **A**

6. **Open Concept:** The double helix is held together by non-covalent hydrogen bonds between complementary nitrogenous bases (A-T have 2 bonds; G-C have 3). Heating (denaturation) breaks these weak bonds, separating the strands without breaking the covalent phosphodiester backbone. Cooling allows complementary strands to re-anneal.

7. **E**

8. **True**

9. **B**

10. **Spliceosome**

11. **C**

12. **Matching:** 1-D, 2-B, 3-C, 4-A

13. **A**

14. **True**

15. **D**

16. **Restriction (or START)**

17. **B** (Sister chromatids separate in both Anaphase II and Mitosis; condensation, alignment, and envelope breakdown occur in all three. Crossing over is exclusive to Meiosis I Prophase).

18. **Open Concept:** The intrinsic pathway is triggered by intracellular stress (e.g., DNA damage), releasing cytochrome c from mitochondria to form the apoptosome, activating initiating **Caspase-9**. The extrinsic pathway is triggered by extracellular death ligands (e.g., FasL) binding to death receptors, activating initiating **Caspase-8**. Both converge on executioner caspases like Caspase-3.

19. **E**

20. **True**

21. **A**

22. **Nucleus**

23. **C**

24. **True**

25. **D**

26. **Open Concept:** A proto-oncogene normally promotes cell growth; a gain-of-function mutation converts it into an oncogene, driving unchecked proliferation. A tumor suppressor gene normally inhibits the cell cycle or promotes apoptosis; a loss-of-function mutation removes these brakes, allowing cancer to progress.

27. **Apoptosome**

28. **E**

29. **False** (Clathrin-mediated endocytosis is highly specific and receptor-mediated, unlike bulk pinocytosis).

30. **B**

Part II: Histology

31. **C**

32. **Tight**

33. **E**

34. **True**

35. **B**

36. **Open Concept:** The ECM of connective tissue proper consists of fibrous proteins (primarily collagen type I for tensile strength, and elastin for elasticity) and ground substance (a highly hydrated gel made of glycosaminoglycans (GAGs), proteoglycans, and multiadhesive glycoproteins like fibronectin), which resists compressive forces.

37. **A**

38. **Chondrocytes**

39. **B** (Volkmann's canals connect Haversian canals to each other and the periosteum; canaliculi connect osteocyte lacunae).

40. **D**

41. **C**

42. **Sarcomere**

43. **D**

44. **False** (Schwann cells form myelin in the PNS; oligodendrocytes form myelin in the CNS).

45. **A**

46. **Matching:** 1-D, 2-B, 3-A, 4-C

47. **D**

48. **Intramembranous**

49. **B**

50. **False** (Intercalated discs are specific to cardiac muscle, which is involuntary).

51. **C**

52. **Open Concept:** The neuromuscular junction consists of a motor neuron axon terminal, a synaptic cleft, and the folded sarcolemma of the muscle fiber (motor endplate). Action potentials trigger calcium influx, causing exocytosis of acetylcholine (ACh) vesicles. ACh crosses the cleft, binds nicotinic receptors on the sarcolemma, initiating muscle depolarization.

53. **Fibroblast**

54. **E**

Part III: Embryology

55. **C**

56. **A** (The nucleus pulposus of the intervertebral disc is an embryonic derivative of the notochord).

57. **Sertoli**

58. **False** (Implantation begins around day 6-7).

59. **D**

60. **Open Concept:** The placental barrier facilitates nutrient/gas exchange and waste removal while preventing immune rejection. It is formed by fetal tissues (syncytiotrophoblast, cytotrophoblast, extraembryonic mesoderm, and fetal capillary endothelium). Maternal blood bathes the villi in intervillous spaces but normally does not mix directly with fetal blood.

61. **B**

62. **Neurulation**

63. **E**

64. **True**

65. **C**

66. **Matching:** 1-D, 2-B, 3-C, 4-A

Part IV: Interdisciplinary

67. **B**

68. **True or False Cluster:**

- A) **True**
- B) **True**
- C) **True**
- D) **False** (Mitochondria can be detected by specific stains like iron hematoxylin or enzyme histochemistry).

69. **Open Concept:** In Sickle Cell Disease, *ex vivo* CRISPR-Cas9 editing targets patient hematopoietic stem cells (CD34+). Cas9 induces a double-strand break at the *BCL11A* enhancer, repaired by error-prone Non-Homologous End Joining (NHEJ), which disrupts the repressor. This de-represses fetal hemoglobin (HbF) gene expression. Histologically, high HbF prevents the pathological polymerization of mutated sickle hemoglobin (HbS) under hypoxia, allowing erythrocytes to maintain their flexible, biconcave disc morphology, thereby preventing microvascular occlusion.

70. **Matching:** 1-C, 2-D, 3-B, 4-A